

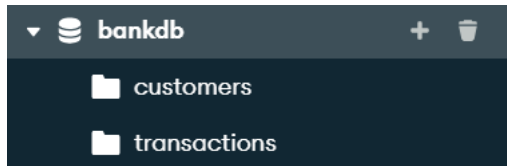
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

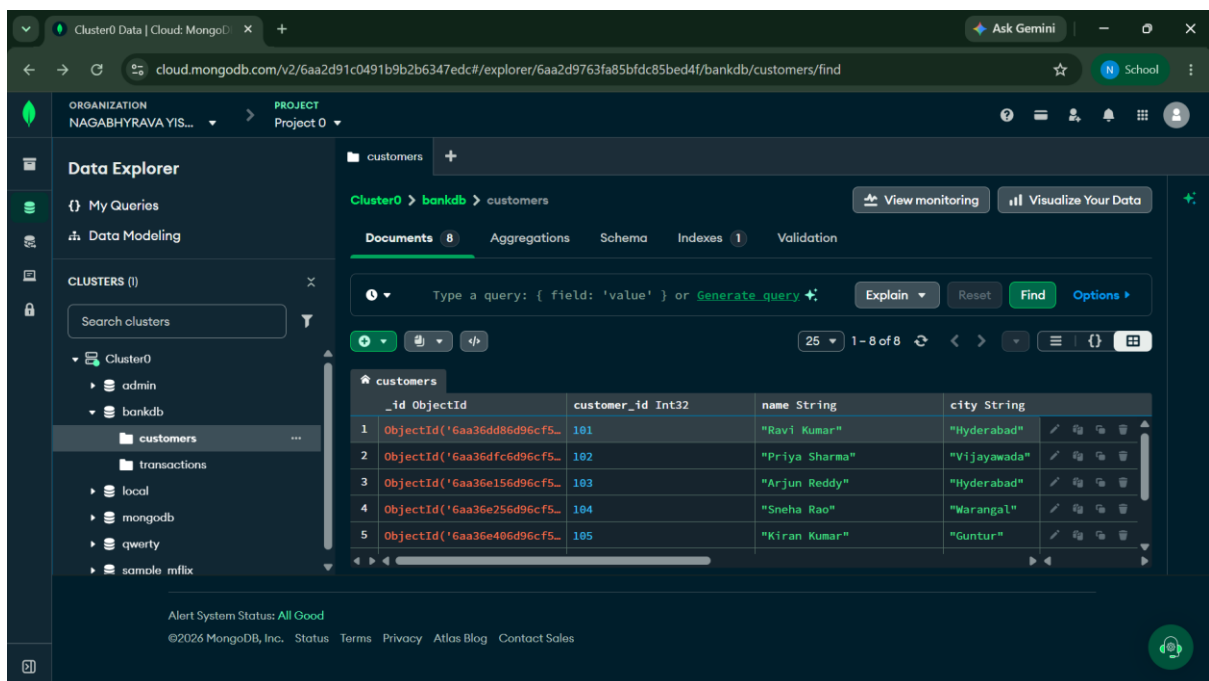
ROLL NO: 2520030414

Name: NAGABHYRAVA YISHNAANI

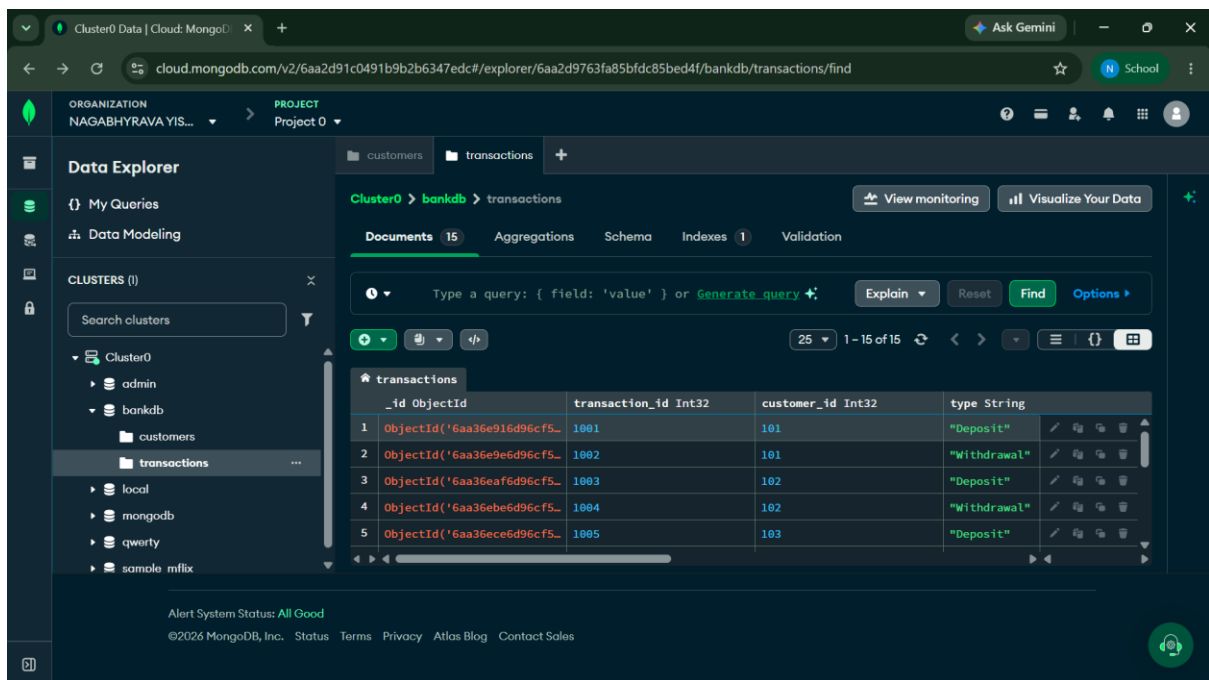
Create database



Insert into the customers

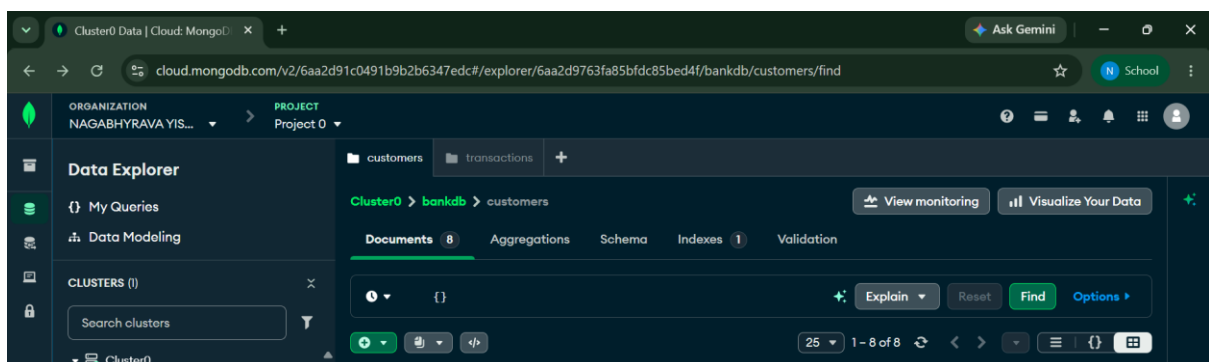


Insert into the transactions

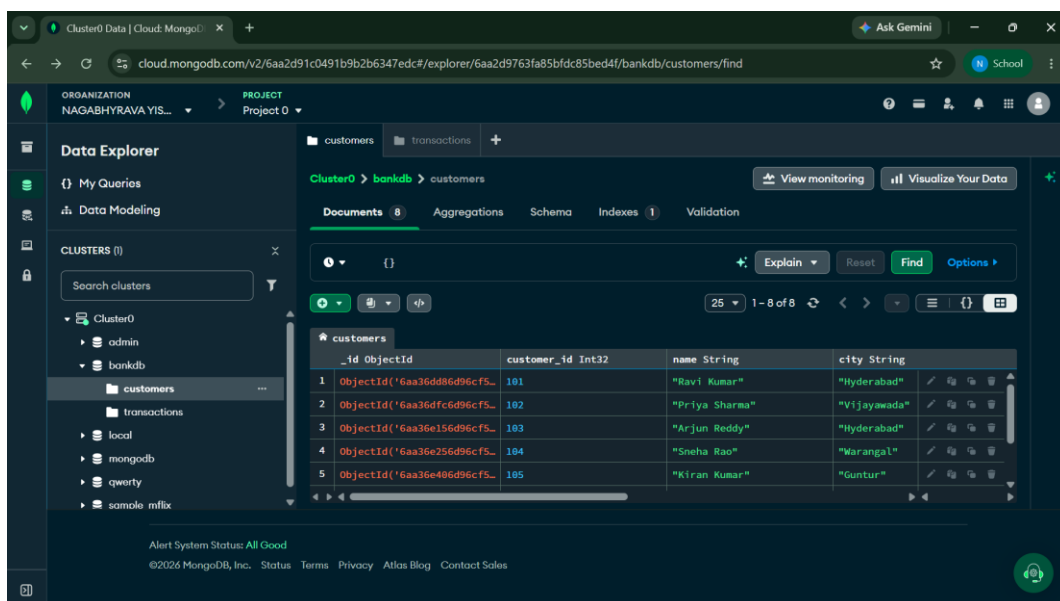


QUERY 1: DISPLAY ALL CUSTOMERS

Query:

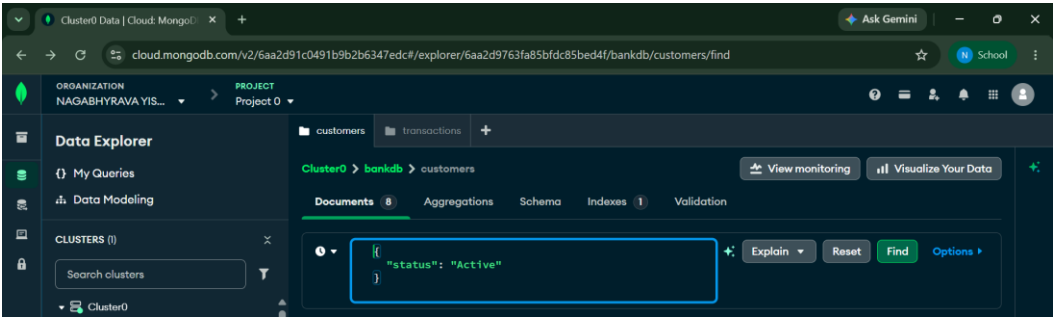


Output:

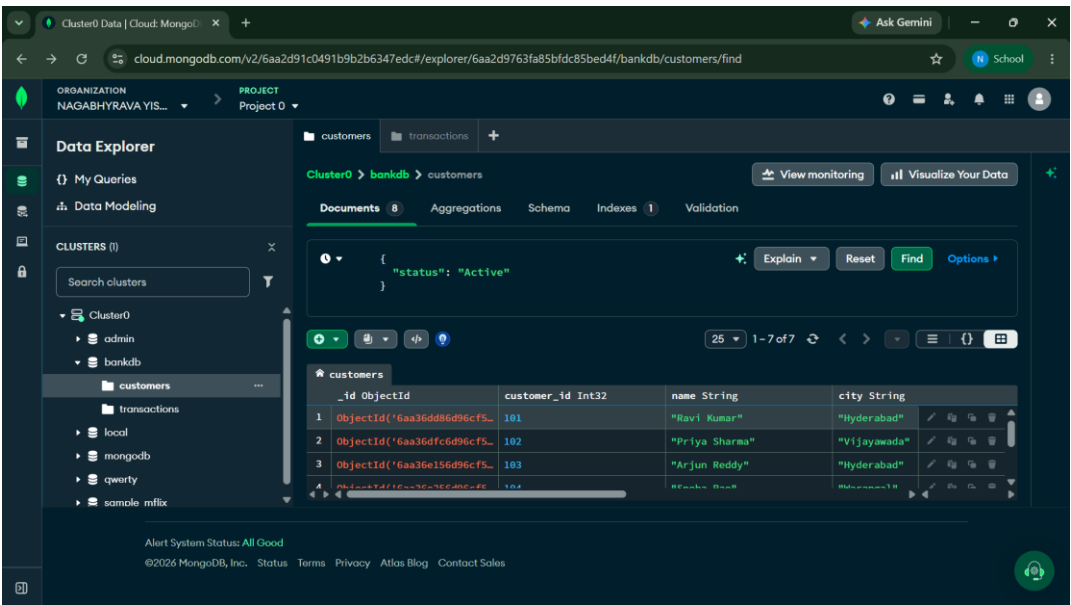


QUERY 2: FIND ACTIVE CUSTOMERS

Query:

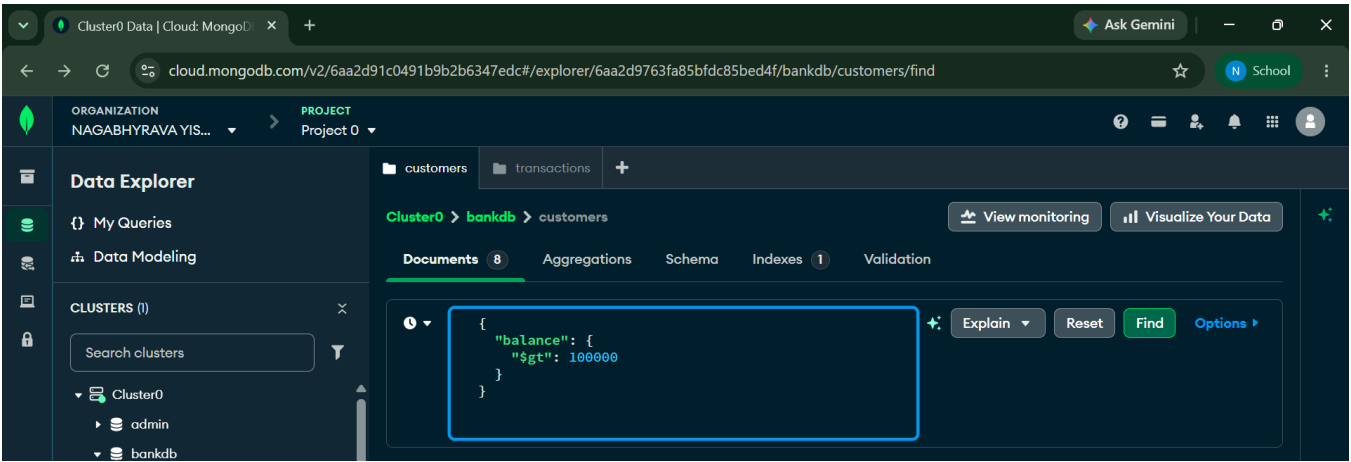


Output:



QUERY 3: FIND CUSTOMERS WITH BALANCE GREATER THAN 100000

Query:



Output:

Cluster0 Data | Cloud: Mongo | Ask Gemini

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bfc85bed4f/bankdb/customers/find

ORGANIZATION NAGABHYRAVA YIS... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- local
- mongodb
- qwerty
- sample mflix

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

{ "balance": { "\$gt": 100000 } }

25 1 - 4 of 4

	_id ObjectId	customer_id Int32	name String	city String
1	ObjectId('6aa36dfc6d96cf5...')	102	Priya Sharma	Vijayawada
2	ObjectId('6aa36e256d96cf5...')	104	Sneha Rao	Warangal
3	ObjectId('6aa36e506d96cf5...')	106	Anjali Singh	Hyderabad
4	ObjectId('6aa36e7c6d96cf5...')	108	Meena Devi	Chennai

Alert System Status: All Good

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QUERY 4: FIND HYDERABAD CUSTOMERS

Query:

Cluster0 Data | Cloud: Mongo | Ask Gemini

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bfc85bed4f/bankdb/customers/find

ORGANIZATION NAGABHYRAVA YIS... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- local
- mongodb
- qwerty
- sample mflix

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

{ "city": "Hyderabad" }

25 1 - 3 of 3

	_id ObjectId	customer_id Int32	name String	city String
1	ObjectId('6aa36d86d96cf5...')	101	Ravi Kumar	Hyderabad
2	ObjectId('6aa36e156d96cf5...')	103	Arjun Reddy	Hyderabad
3	ObjectId('6aa36e506d96cf5...')	106	Anjali Singh	Hyderabad

Alert System Status: All Good

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OUTPUT:

Cluster0 Data | Cloud: Mongo | Ask Gemini

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bfc85bed4f/bankdb/customers/find

ORGANIZATION NAGABHYRAVA YIS... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- local
- mongodb
- qwerty
- sample mflix

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

{ "city": "Hyderabad" }

25 1 - 3 of 3

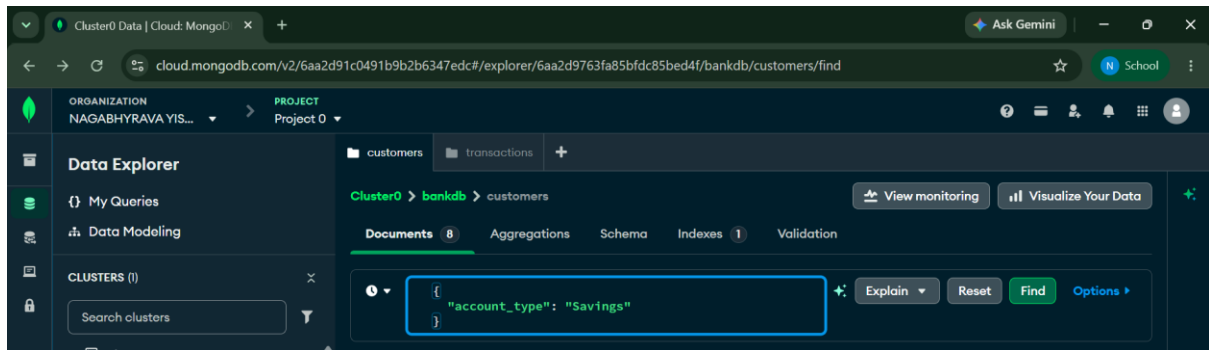
	_id ObjectId	customer_id Int32	name String	city String
1	ObjectId('6aa36d86d96cf5...')	101	Ravi Kumar	Hyderabad
2	ObjectId('6aa36e156d96cf5...')	103	Arjun Reddy	Hyderabad
3	ObjectId('6aa36e506d96cf5...')	106	Anjali Singh	Hyderabad

Alert System Status: All Good

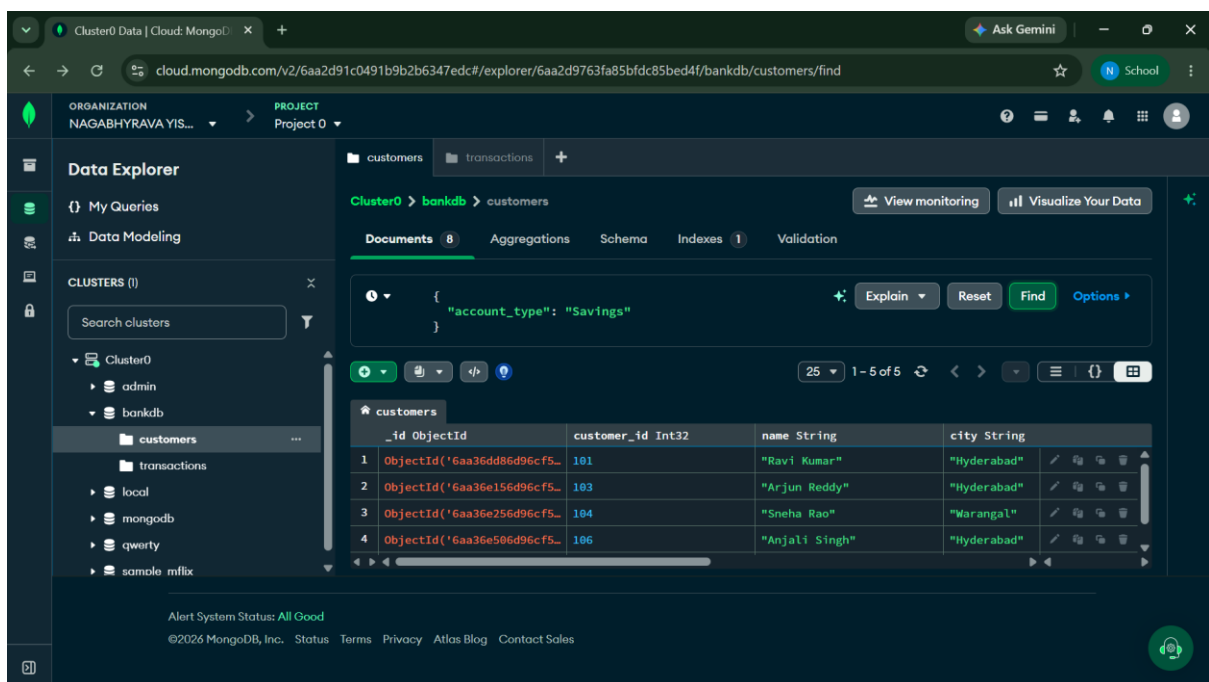
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QUERY 5: FIND SAVINGS ACCOUNT CUSTOMERS

Query:

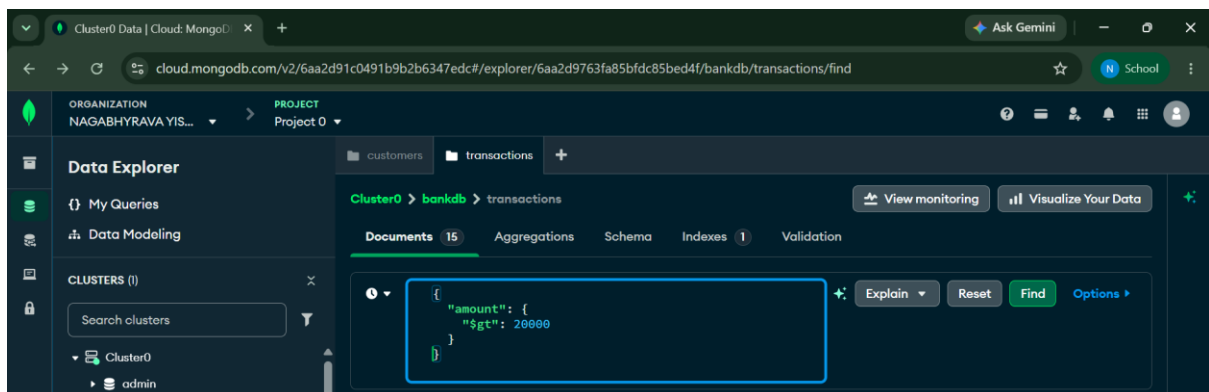


Output:



QUERY 6: FIND TRANSACTIONS GREATER THAN 20000

Query:



Output:

Cluster0 Data | Cloud: Mongo

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bdc85bed4f/bankdb/transactions/find

ORGANIZATION NAGABHYRAVA YIS... PROJECT Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- local
- mongodb
- qwerty
- sample mflix

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

{ "amount": { "\$gt": 20000 } }

25 1-9 of 9

	_id ObjectId	transaction_id Int32	customer_id Int32	type String
1	ObjectId('6aa36e916d96cf5...')	1001	101	"Deposit"
2	ObjectId('6aa36eaf6d96cf5...')	1003	102	"Deposit"
3	ObjectId('6aa36ece6d96cf5...')	1005	103	"Deposit"
4	ObjectId('6aa36eeb6d96cf5...')	1007	104	"Deposit"
5	ObjectId('6aa36ef96d96cf5...')	1008	104	"Withdrawal"

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QUERY 7: FIND DEPOSITS

Query:

Cluster0 Data | Cloud: Mongo

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bdc85bed4f/bankdb/transactions/find

ORGANIZATION NAGABHYRAVA YIS... PROJECT Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

{ "type": "Deposit" }

Output:

Cluster0 Data | Cloud: Mongo... | Ask Gemini | School

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bdc85bed4f/bankdb/transactions/find

ORGANIZATION NAGABHYRAVA YIS... PROJECT Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- local
- mongodb
- qwerty
- sample mflix

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

Find

transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String
1	ObjectId('6aa36e916d96cf5...')	1001	101	"Deposit"
2	ObjectId('6aa36eaf6d96cf5...')	1003	102	"Deposit"
3	ObjectId('6aa36ece6d96cf5...')	1005	103	"Deposit"
4	ObjectId('6aa36eeb6d96cf5...')	1007	104	"Deposit"

Alert System Status: All Good

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QUERY 8: FIND ATM TRANSACTIONS

Query:

Cluster0 Data | Cloud: Mongo... | Ask Gemini | School

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bdc85bed4f/bankdb/transactions/find

ORGANIZATION NAGABHYRAVA YIS... PROJECT Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

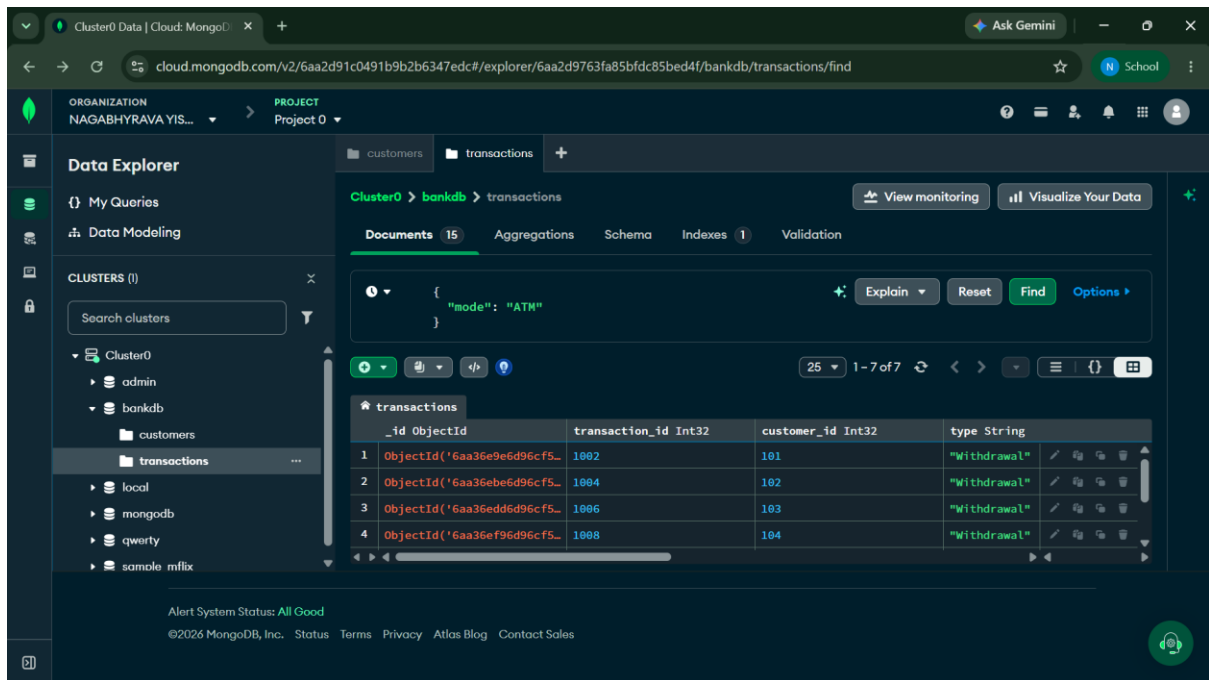
View monitoring Visualize Your Data

Find

transactions

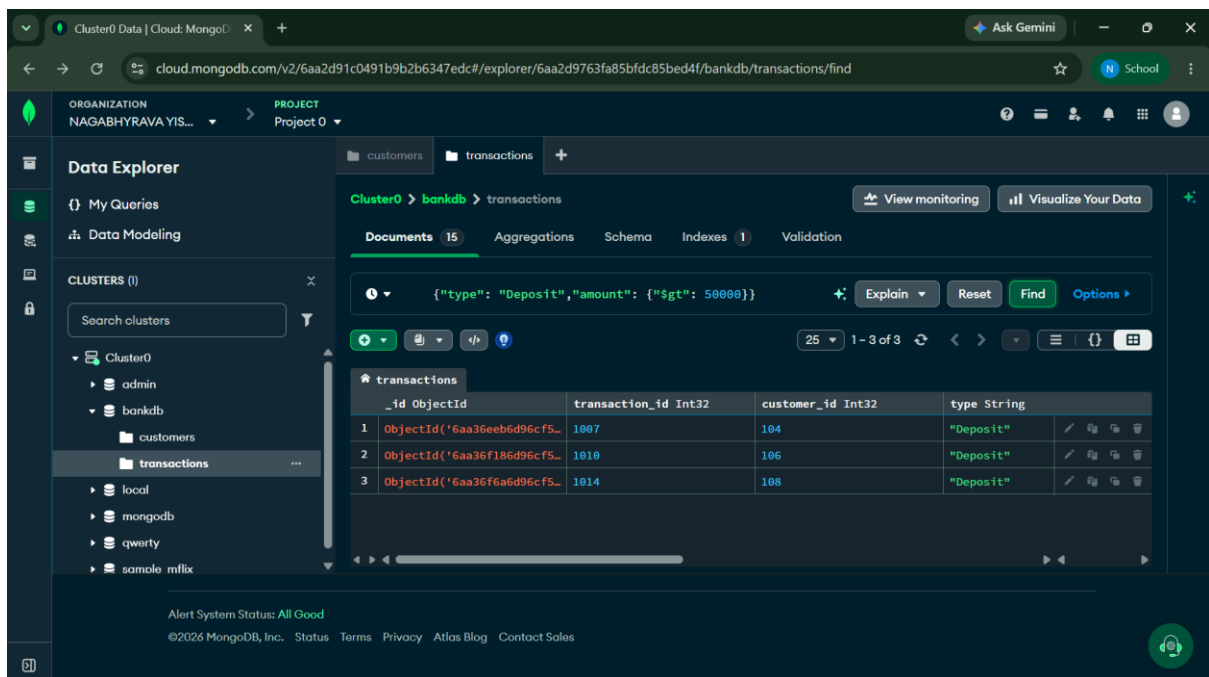
```
{  
  "mode": "ATM"  
}
```

Output:



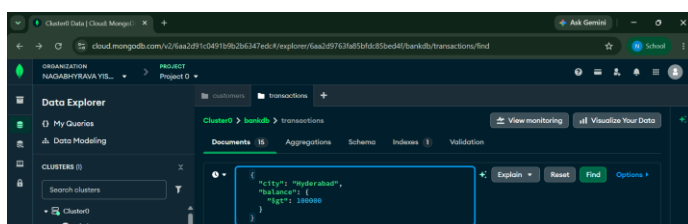
QUERY 9: FIND DEPOSITS GREATER THAN 50000

Query + output:



QUERY 10: FIND HYDERABAD CUSTOMERS WITH HIGH BALANCE

Query:



Output:

The screenshot shows the MongoDB Cloud Data Explorer interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters and databases. The main panel shows the 'customers' collection in the 'bankdb' database. A query is executed, returning a single document with the following structure:

```
{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}
```

Below the query result, a table view shows the document details:

_id	customer_id	name	city
1	ObjectId('6aa36e596d96cf5...')	Anjali Singh	Hyderabad

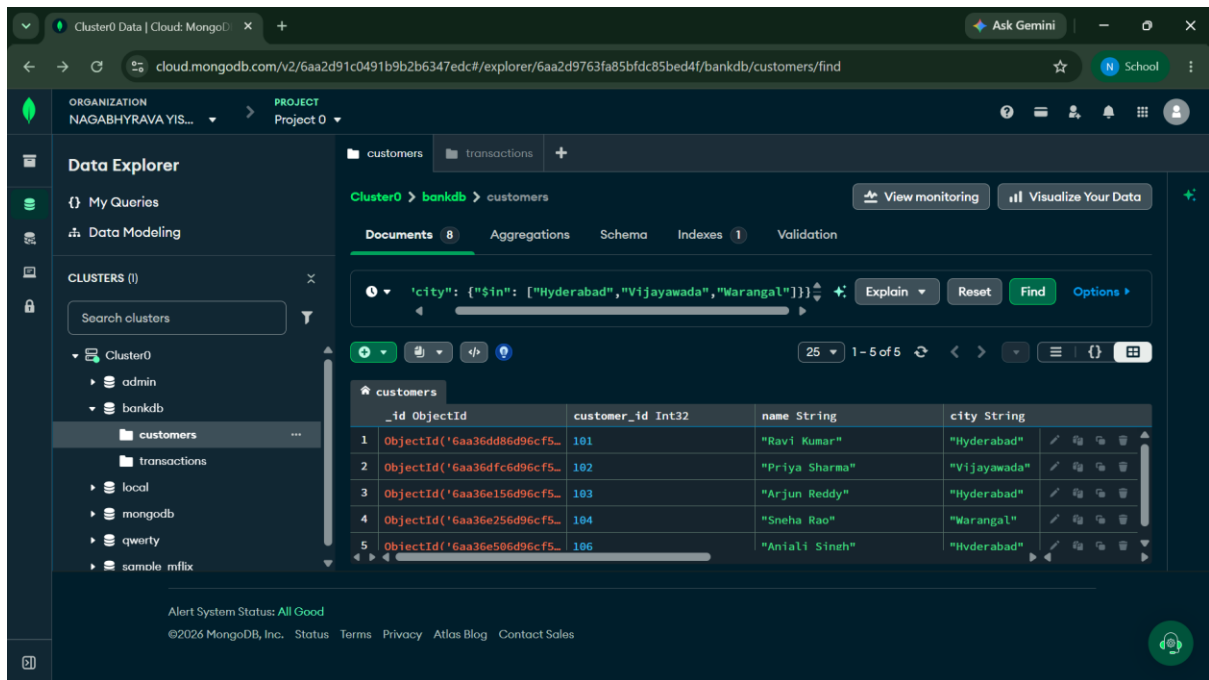
QUERY 11: FIND CUSTOMERS IN MULTIPLE CITIES

Query:

The screenshot shows the MongoDB Cloud Data Explorer interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters and databases. The main panel shows the 'customers' collection in the 'bankdb' database. A query is entered in the query editor:

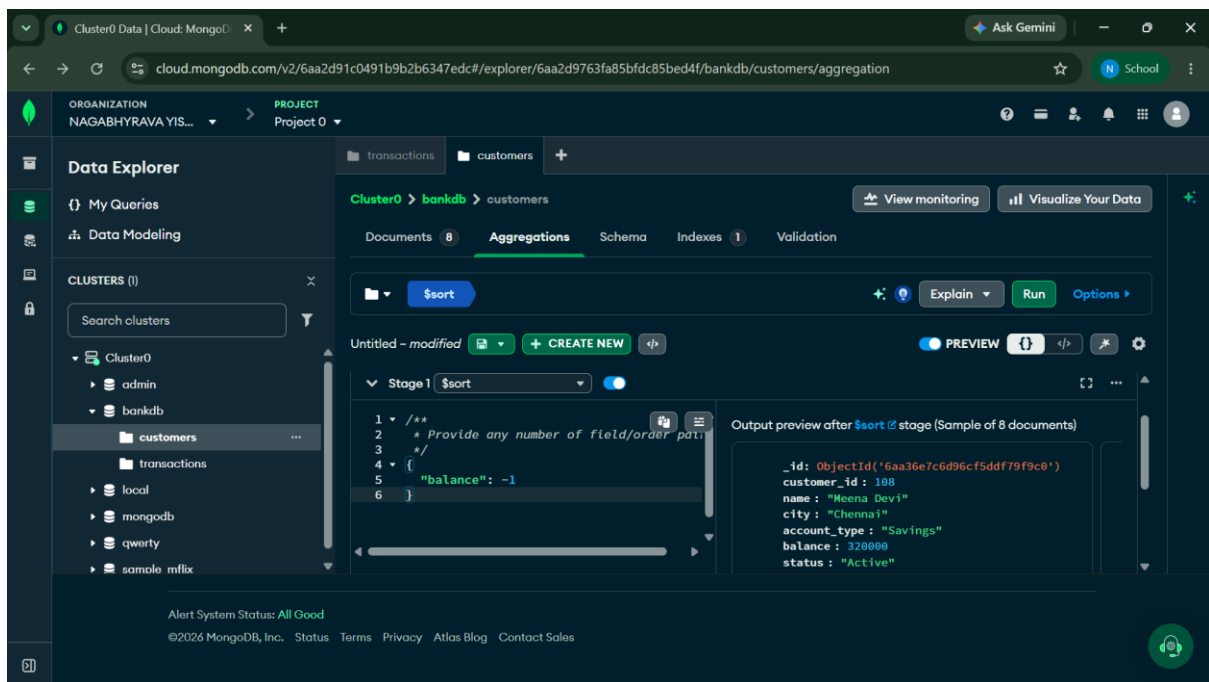
```
{
  "city": {
    "$in": [
      "Hyderabad",
      "Vijayawada",
      "Warangal"
    ]
  }
}
```

Output:



QUERY 12: SORT CUSTOMERS BY BALANCE

Query:



Output:

```

_id: ObjectId('6aa36d...
customer_id : 101
name : "Ravi Kumar"
city : "Hyderabad"
account_type : "Saving...
balance : 85000
status : "Active"

_id: ObjectId('6aa36d...
customer_id : 102
name : "Priya Sharma"
city : "Vijayawada"
account_type : "Curren...
balance : 125000
status : "Active"

_id: ObjectId('6aa36e...
customer_id : 103
name : "Arjun Reddy"
city : "Hyderabad"
account_type : "Saving...
balance : 45000

```

```

status : "Active"

_id: ObjectId('6aa36e...
customer_id : 104
name : "Sneha Rao"
city : "Warangal"
account_type : "Saving...
balance : 210000
status : "Active"

_id: ObjectId('6aa36e...
customer_id : 105
name : "Kiran Kumar"
city : "Guntur"
account_type : "Curren...
balance : 65000
status : "Inactive"

_id: ObjectId('6aa36e...
customer_id : 106
name : "Anjali Singh"
city : "Hyderabad"

```

```

account_type : "Saving...
balance : 175000
status : "Active"

_id: ObjectId('6aa36e...
customer_id : 107
name : "Rahul Verma"
city : "Bengaluru"
account_type : "Curren...
balance : 95000
status : "Active"

_id: ObjectId('6aa36e...
customer_id : 108
name : "Meena Devi"
city : "Chennai"
account_type : "Saving...
balance : 320000
status : "Active"

```

QUERY 13: CALCULATE TOTAL TRANSACTION AMOUNT

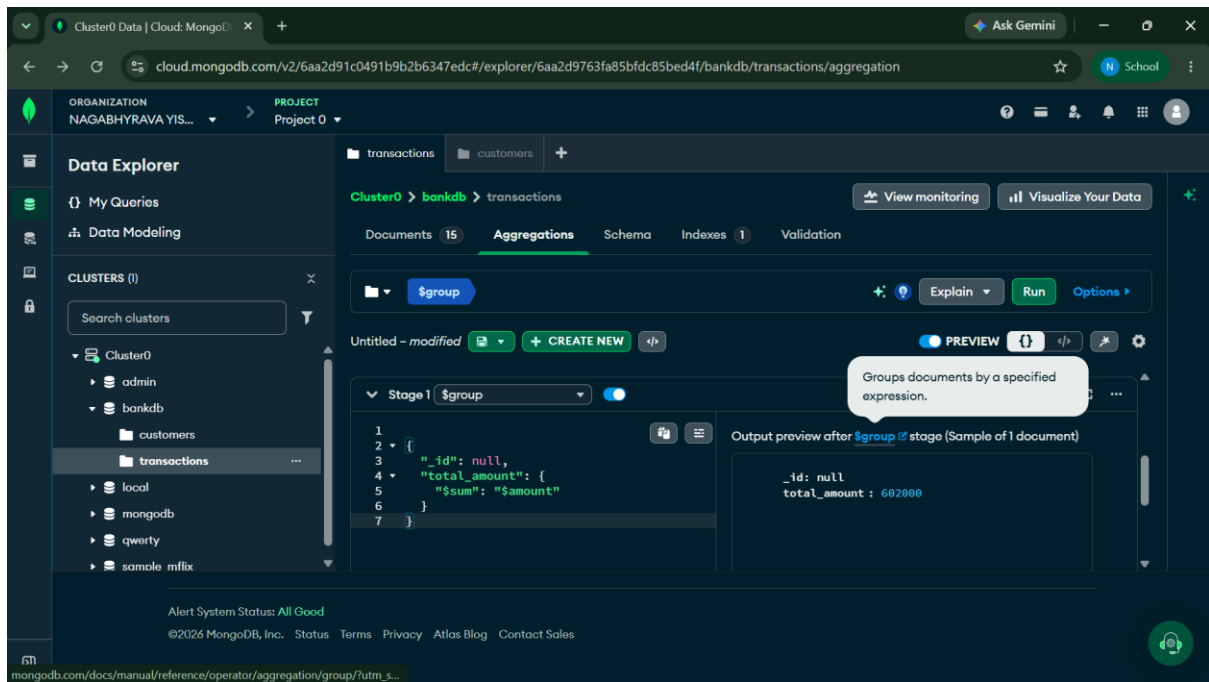
Query:

```

Stage 1 $group
1
2 {
3   "_id": null,
4   "total_amount": {
5     "$sum": "$amount"
6   }
7 }

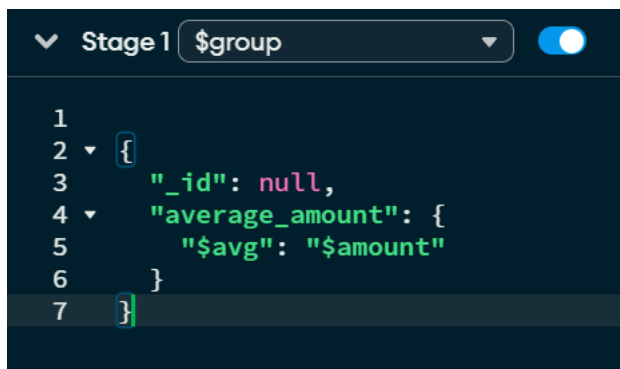
```

Output:

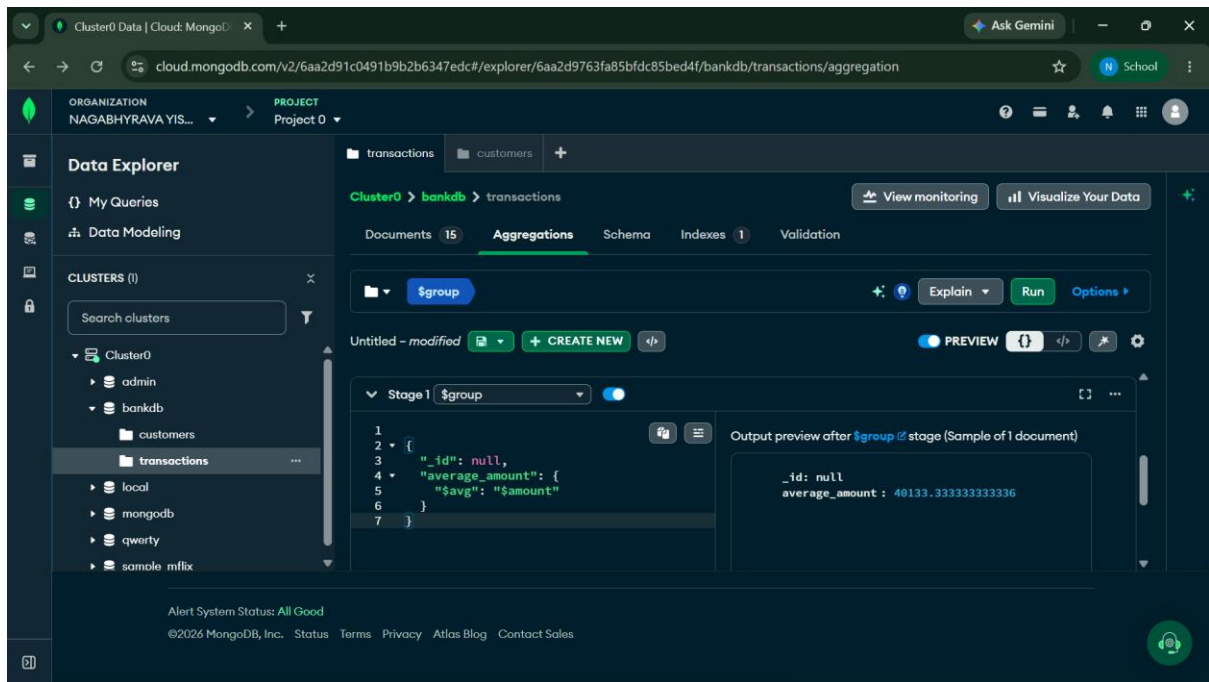


QUERY 14: CALCULATE AVERAGE TRANSACTION AMOUNT

Query:

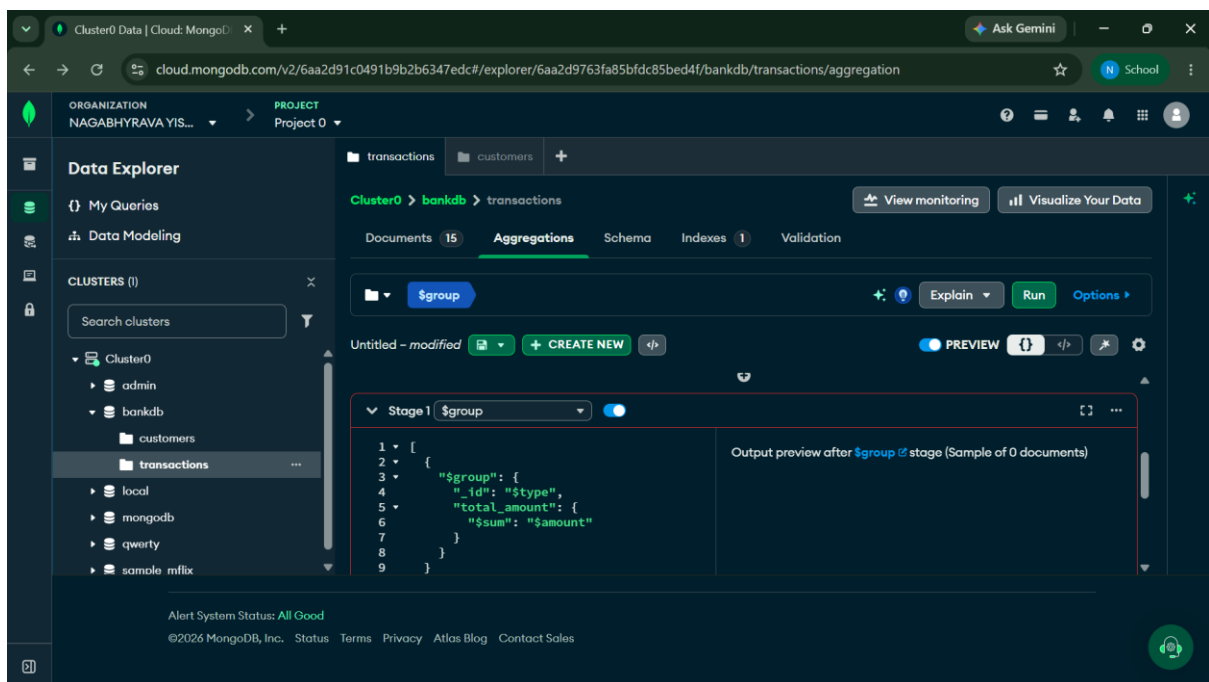


Output:



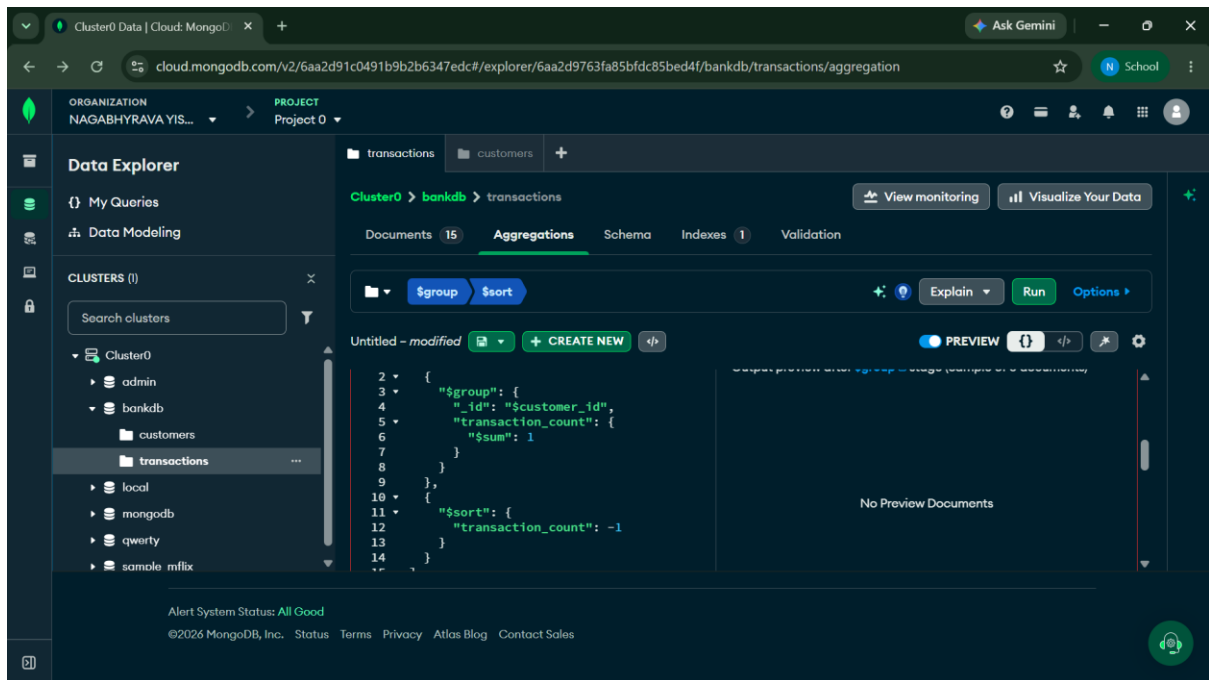
QUERY 15: TOTAL TRANSACTION AMOUNT BY TYPE

Query + output:



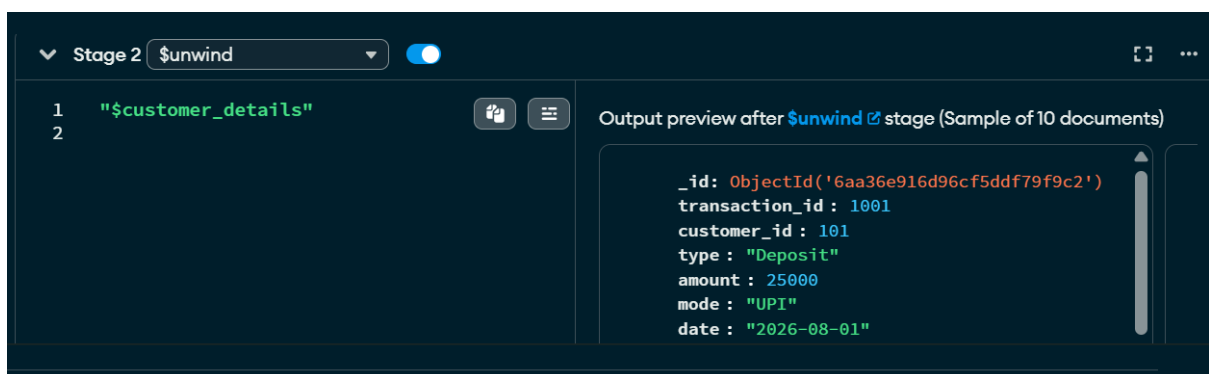
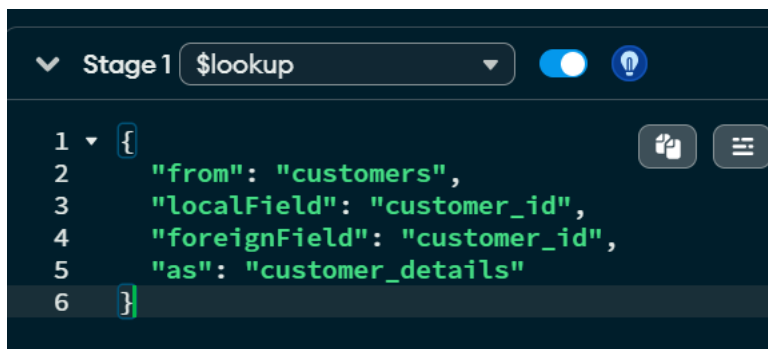
QUERY 16: NUMBER OF TRANSACTIONS PER CUSTOMER

Query + output:



QUERY 17: CUSTOMER + TRANSACTION JOIN

Query:



OUTPUT:

Cluster0 Data | Cloud: Mongo | Ask Gemini | School

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85b5fdc85bed4f/bankdb/transactions/aggregation

ORGANIZATION PROJECT

Stage 1: \$lookup ENABLED ADD STAGE

STAGE INPUT Sample of 10 documents

```

1 {
2   from: "customers",
3   localField: "customer_id",
4   foreignField: "customer_id",
5   as: "customer_details"
6 }

```

STAGE OUTPUT Sample of 10 documents

```

1 {
2   date: "2026-08-01",
3   status: "Success",
4   customer_details: {
5     _id: ObjectId('6aa36e...'),
6     customer_id: 101,
7     name: "Ravi Kumar",
8     city: "Hyderabad",
9     account_type: "Savings",
10    balance: 85000,
11    status: "Active"
12   }
13 }

```

Cluster0 Data | Cloud: Mongo | Ask Gemini | School

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85b5fdc85bed4f/bankdb/transactions/aggregation

ORGANIZATION PROJECT

Stage 2: \$unwind ENABLED ADD STAGE

STAGE INPUT Sample of 10 documents

```

1 "$customer_details"
2

```

STAGE OUTPUT Sample of 10 documents

```

1 {
2   date: "2026-08-01",
3   status: "Success",
4   customer_details: {
5     _id: ObjectId('6aa36e...'),
6     customer_id: 101,
7     name: "Ravi Kumar",
8     city: "Hyderabad",
9     account_type: "Savings",
10    balance: 85000,
11    status: "Active"
12   }
13 }

```

QUERY 18: DISPLAY CUSTOMER NAME AND TRANSACTION AMOUNT

Query:

Stage 1 \$lookup

```
1 {
2   "from": "customers",
3   "localField": "customer_id",
4   "foreignField": "customer_id",
5   "as": "customer_details"
6 }
```

Stage 2 \$unwind

```
1 "$customer_details"
2
```

Output preview after \$unwind stage (Sample of 10 documents)

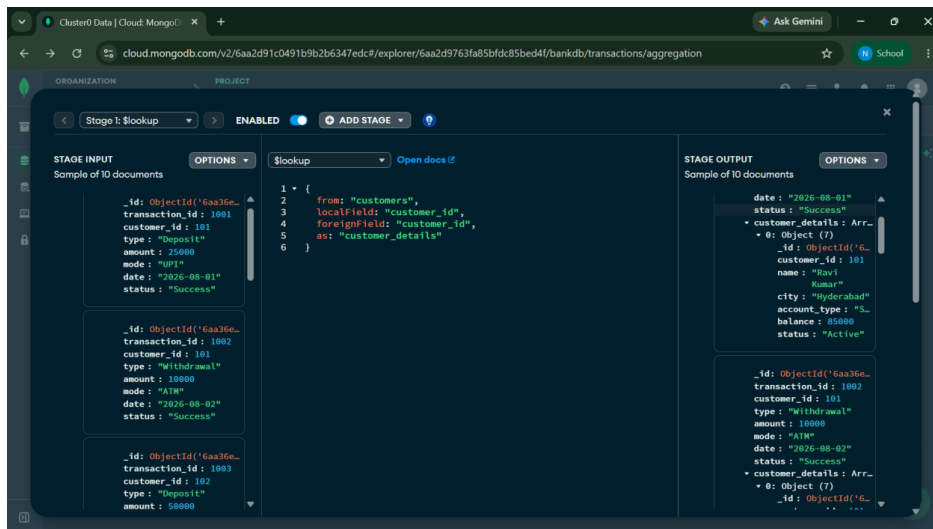
```
_id: ObjectId('6aa36e916d96cf5ddf79f9c2')
transaction_id: 1001
customer_id: 101
type: "Deposit"
amount: 25000
mode: "UPI"
date: "2026-08-01"
```

\$lookup \$unwind \$project

Untitled - modified

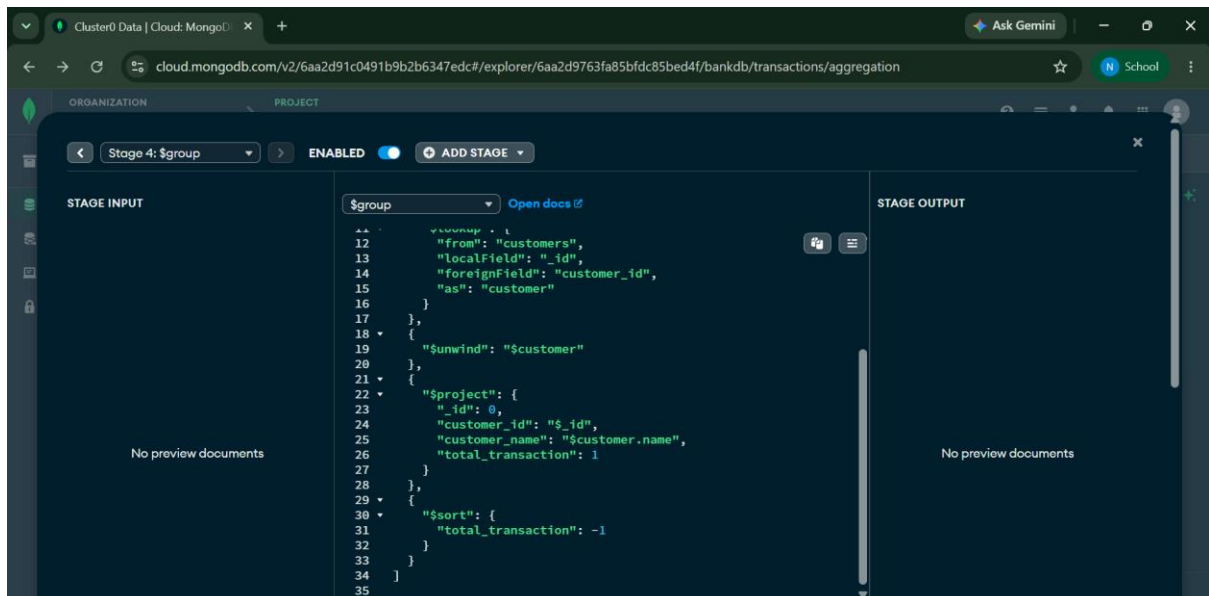
```
1 {
2   "$project": {
3     "_id": 0,
4     "transaction_id": 1,
5     "customer_name": "$customer.name",
6     "city": "$customer.city",
7     "type": 1,
8     "amount": 1,
9     "mode": 1
10  }
11 }
12
13
```

Output:



QUERY 19: TOTAL TRANSACTION AMOUNT FOR EACH CUSTOMER

Query:



QUERY 20: COMPLEX BANK TRANSACTION REPORT

Query + output:

Cluster0 Data | Cloud: Mongo

Ask Gemini

cloud.mongodb.com/v2/6aa2d91c0491b9b2b6347edc#/explorer/6aa2d9763fa85bdc85bed4f/bankdb/transactions/aggregation

School

ORGANIZATION

PROJECT

Stage 4: \$group

ENABLED

ADD STAGE

STAGE INPUT

No preview documents

\$group

Open docs

```
9  },
10 {
11   "$lookup": {
12     "from": "transactions",
13     "localField": "customer_id",
14     "foreignField": "customer_id",
15     "as": "transactions"
16   },
17 },
18 {
19   "$unwind": "$transactions"
20 },
21 {
22   "$group": {
23     "_id": "$customer_id",
24     "customer_name": {
25       "$first": "$name"
26     },
27     "city": {
28       "$first": "$city"
29     },
30     "balance": {
31       "$first": "$balance"
32     },
33     "total_transactions": {
34       "$sum": "$transactions.amount"
35     },
36     "transaction_count": {
```

STAGE OUTPUT

No preview documents